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Position-based visual servo control of autonomous robotic manipulators

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ABSTRACT

This paper concerns the position-based visual servo control of autonomous robotic manipulators in space. It focuses on the development of a real-time vision-based pose and motion estimation algorithm of a non-cooperative target by photogrammetry and extended Kalman filter for robotic manipulators to perform autonomous capture. Optical flow algorithm is adopted to track the target features in order to improve the image processing efficiency. Then, a close-loop position-based visual servo control strategy is devised to determine the desired pose of the end-effector at the rendezvous point based on the estimated pose and motion of the target. The corresponding desired joint angles of the robotic manipulator in the joint space are derived by the inverse kinematics of the robotic manipulator. The developed algorithm and position-based visual servo control strategy are validated experimentally by a custom built robotic manipulator with an eye-in-hand configuration. The experimental results demonstrate the proposed estimation algorithm and control scheme are feasible and effective.

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1. Introduction

Robotic manipulators have been widely used in space for docking, assembling, repairing and other on-orbit servicing operations [1–5]. For instance, Mobile Servicing System (MSS) or Canadarm2 [6], Japanese Experiment Module Remote Manipulator System (JEMRMS) [7] and European Robotic Arm (ERA) [8] are typical examples of robotic manipulators performing assembly, maintenance, and payloads exchanging tasks on International Space Station. These operations were conducted either autonomously or by human astronauts. Robotic manipulators mounted on Mars exploration rovers, such as, Viking 1 and 2 [9], Spirits and Opportunity [10], Phoenix [11] and Curiosity [12], were designed to collect soil samples and/or place instruments on a target. These tasks were performed by preprogrammed commands and controlled from the Earth directly or relayed by the Mars Orbiter. Cameras were used in these missions to monitor the movements of the manipulators and take photographs of the surroundings. Robotic manipulators of orbital docking systems, such as the Shuttle Remote Manipulator System [13] and Orbital Express [14], performed tasks of grappling, docking, refueling, repairing and/or servicing another spacecraft. Pure experimental systems, such as, ROTEX (Robot Technology Experiment) and ETS-VII (Engineering Test Satellite) [15] demonstrated the operations of assembling, grasping, docking and exchanging orbit replaceable units by robotic manipulators. Most of these missions employed human-in-the-loop control. Manual control from the Earth may result in long time delay, while sending astronauts into space to perform the tasks suffers higher cost and the possibility of life loss. To address these challenges, autonomous control is required and becomes a research highlight in the field of robotic technology [16,17].

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Nomenclature

$\boldsymbol{A}$	system transformation matrix
$\boldsymbol{B}$	process noise coefficient matrix
$\boldsymbol{E}$	residual error in the measurement
f	focal length of the camera
$\boldsymbol{H}$	Jacobian matrix of measurement model
$\boldsymbol{I}$	identity matrix
$\boldsymbol{J_c}$	Jacobian matrix of the pin-hole camera model
$\boldsymbol{J_r}$	Jacobian matrix of the robotic manipulator
$\boldsymbol{P}$	covariance matrix of system state variable
$\boldsymbol{Q}$	covariance matrix of process noise
$\boldsymbol{R}$	covariance matrix of measurement noise
t	sample time
$\boldsymbol{X}$	state variable
$\boldsymbol{Z}$	measurement vector
$(\cdot)^{\boldsymbol{c}}$	control vector
$(\cdot)^{\boldsymbol{d}}$	desired vector
$(\cdot)^{\boldsymbol{t}}$	vector of the target
$(\bullet)_C$	coordinates in camera frame
$(\bullet)_m$	image coordinates
$(\bullet)_T$	coordinates in target frame
$(\bullet)_{To}$	coordinates of target frame origin in camera frame
d_6	variable of robotic gripper
$\boldsymbol{dX_p}$	correction term of the photogrammetry
$\boldsymbol{K_g}$	Kalman gain
$\boldsymbol{R_{TC}}$	rotational matrix from target frame to camera frame
$\boldsymbol{R_{CT}}$	rotational matrix from camera frame to target frame
$\boldsymbol{T_{cg}}$	transformation matrix from camera frame to global frame
$\boldsymbol{X_c}$	position of the grasp point in camera frame
$\boldsymbol{X_g}$	position of the grasp point in global frame
$\boldsymbol{X_p}$	photogrammetry estimated target pose in camera frame
$\boldsymbol{\delta}$	vector of image error
ε	pre-set tolerance of photogrammetry
$\boldsymbol{\mu}$	measurement noise vector
$\boldsymbol{\omega}$	process noise vector
$\boldsymbol{\Theta}$	main robotic joint angle vector
ΔT	estimated time for capture
$\theta_{1,2,3,4,5}$	angle of torso, shoulder, elbow, wrist roll and wrist yaw joints
$\theta_{x,y,z}$	target orientation refer to x, y, z axes of camera frame

Autonomous control of robotic manipulator to track and grasp a moving target requires the precise knowledge of the target's pose and motion. Because of the non-intrusive, non-damaging and non-contact nature, computer vision is favored exclusively as a sensing system to obtain the required information [2,16,18–22]. Accordingly, visual servo control system has been developed to control the pose of manipulator's end-effector with respect to the target based on the feedback of vision system. For instance, the position of a known moving object in the image plane can be tracked with a single mobile camera based on past images and past control inputs to the mobile platform [23]. The autonomous capture of a non-cooperative target by a robotic manipulator requires not only to track the motion of target [24,25] but also to predict the rendezvous point and follow a specific approaching trajectory by the end-effector based on the estimated pose and motion of the target [16,19].

The camera configuration in a visual servo robotic system can be either eye-in-hand or eye-to-hand [26]. The eye-in-hand camera is mounted on the end-effector to provide a close and precise view of the target while the eye-to-hand camera is installed beside the robot to monitor the whole workspace with a broad and relative less accurate scene of the target [27]. Based on the errors employed in control, the robotic visual servo may be categorized as: image-based, position-based, and hybrid visual servo [28,29]. The image-based visual servo (IBVS) controls robots by the error between the projected desired and actual positions in the 2D (two dimensional) image plane via an image Jacobian without reconstruction of the target. Thus, it is free from target model errors and less sensitive to camera calibration errors and measurement noise in images. Considerable efforts [30,31] have been devoted to track moving targets in 3D (three dimensional) space with eye-in-hand cameras using IBVS. Extended Kalman filter was introduced into the IBVS algorithm to address the navigation errors and actuation delays [32]. The perturbation to eye-in-hand cameras by the flexibility of robotic manipulator [33] was investigated to enhance the robustness of IBVS algorithm. However, the IBVS lacks 3D depth information of a target and additional measure is required to estimate the depth. The position-based visual servo (PBVS) controls the error between the desired and actual poses and motion of the end-effector directly in the 3D workspace. The advantage of the PBVS is that the pose of end-effector can be controlled relative to the target directly and naturally, while the drawbacks are that the pose and motion estimation is prone to camera calibration errors, target model accuracy, and image measurement noise. These challenges have been successfully addressed by many researchers to eliminate image errors caused by an uncalibrated camera [34,35] and suppress the image noise due to the vibration of camera resulting from flexible manipulators [36]. Finally, the hybrid visual servo, referred as 2½D visual servo in the literature, evaluates the control errors partially in the 3D workspace and partially on the 2D image plane. Although effective, the hybrid system is generally more complex than either IBVS or PBVS for implementation. In the current work, we adopted a single, calibrated and eye-in-hand camera with PBVS to simplify the system configuration and implementation in autonomous capture of non-cooperative targets.

The key issue in the autonomous capture of non-cooperative targets by PBVS robotic manipulator is the estimation of target's pose and motion with visual feedback

to reconstruct the target in 3D space. The pose and motion estimated by the eye-in-hand camera in PBVS are prone to image jittering, residual vibration and unexpected disturbances of camera or the end-effector. Considerable efforts have been devoted to extract information from visual images in the past decades [37–40]. Different methodologies have been developed, such as, analytic or geometric method, learning-based method, offline method and filtering method. The geometric method, such as the photogrammetry, is widely used when the camera is properly calibrated and the target is known. It extracts six degrees of freedom information (pose) of the target from 2D images. However, the geometric method is memoryless and its result is noisy if the image data is not smooth either due to the jittering of image processing or the mechanical disturbance of the end-effector in case of eye-in-hand camera. The learning-based method requires a sufficient large set of images of target in different poses, which is usually not available when the target is non-cooperative. The offline methods are not suitable for the robotic manipulator to track, approach and capture the target in real-time. Another potential solution for the estimation of pose and motion is filtering. Kalman filter (KF) is a widely used filtering algorithm to estimate unknown variables based on a set of noisy measurements observed over time with initial conditions. Since being proposed in 1960 [41], the KF has been applied widely with many variations and extensions beyond the original proposed linear system [42]. Although the initial conditions do not change the convergence property of the KF, they do affect the performance of the filter, especially when dealing with the non-cooperative target in real-time where the initial conditions are unknown.

The focus of this study is to improve the estimation of pose and motion in real-time PBVS control to address the challenges of image jittering and disturbances of camera caused by the flexibility in joints, actuation delays and rough motion of stepper motor. In our previous PBVS work [16], the challenges were studied by a dual KF approach where the first KF was introduced in the image space and the second KF was used in the 3D space. The first KF not only reduces the image noise due to jittering to prevent the errors propagating into photogrammetry algorithm but also provides substitute image data momentarily to enhance the tracking robustness in case of an outrage in the vision system occurs. The second KF was designed to suppress impact of the residual vibration and unexpected distribution of the camera and the sudden motion of a target. Although effective, the approach is computational cumbersome and increases the actuation delay. To address the issue, a new methodology was developed in the current work to integrate the photogrammetry and extended Kalman filter (EKF) to estimate the target's pose and motion in real-time for PBVS. Combining with the inner closed-loop control of robotic manipulator, the visual estimation of pose and motion of a target as well as the control errors in 3D space forms an outer closed-loop control. It is worth pointing out that the effectiveness of EKF in improving the robustness of visual servo [32,33] has also been demonstrated in IBVS. For instance, the EKF introduced in the image space not only improves the estimation accuracy of target's kinematics state but also substitutes state estimates when the target is lost locking [33]. These works showed that the introduction of EKF into IBVS is effective to prevent the tracking failure in a dynamic situation due to image noises and actuation delay. The current approach is validated experimentally by a custom-built robotic manipulator [16] with an eye-in-hand camera mounted closely to the end-effector. The experimental results demonstrate the effectiveness and robustness of the proposed approach by successfully tracking, approaching and capturing a non-cooperative target autonomously.

2. Position-based visual servo of robotic manipulator

2.1. Camera model and photogrammetry

The pose of a target can be described by the Cartesian coordinates $\{x_{To}, y_{To}, z_{To}\}^T$ of a target-fixed frame origin and the Euler angles of that frame $\{\theta_x, \theta_y, \theta_z\}^T$ regarding to the camera frame. The rotational matrix from the camera frame to the target frame, $\boldsymbol{R_{CT}}$, can be developed by rotating x-axis of the camera frame by θ_x first and followed by rotating y-axis of the camera frame by θ_y and rotating z-axis of the camera frame by θ_z. Accordingly, the rotational matrix from the target frame to the camera frame can be expressed as $\boldsymbol{R_{TC}} = \boldsymbol{R_{CT}}^T$, such that

$$\boldsymbol{R_{TC}} = \begin{bmatrix} C_yC_z & -C_yS_z & S_y \\ C_xS_z + S_xS_yC_z & C_xC_z - S_xS_yS_z & -S_xC_y \\ S_xS_z - C_xS_yC_z & S_xC_z + C_xS_yS_z & C_xC_y \end{bmatrix} \tag{1}$$

Here, $S_x = \sin\theta_x$, $S_y = \sin\theta_y$, $S_z = \sin\theta_z$ $C_x = \cos\theta_x, C_y = \cos\theta_y, C_z = \cos\theta_z$.

Assume the coordinates, $\{x_T, y_T, z_T\}^T$, of a feature point on the target are known in the target frame, which implies the vision system is calibrated in advance. Then, the homogeneous relationship between the target and camera frames can be described by

$$\begin{Bmatrix} x_C \\ y_C \\ z_C \\ 1 \end{Bmatrix} = \begin{bmatrix} & & & x_{To} \\ & \boldsymbol{R_{TC}} & & y_{To} \\ & & & z_{To} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} x_T \\ y_T \\ z_T \\ 1 \end{Bmatrix} \tag{2}$$

where $\{x_C, y_C, z_C\}^T$ is the coordinates of the same point in the camera frame.

Consider a pinhole camera model as shown in Fig. 1, denote r_{ij} for elements of $\boldsymbol{R_{TC}}$. The feature point on the target is projected onto the image plane by Eq. (3), such as

$$\begin{cases} x_m = -f\dfrac{x_C}{y_C - f} = -f\dfrac{r_{11}x_T + r_{12}y_T + r_{13}z_T + x_{To}}{r_{21}x_T + r_{22}y_T + r_{23}z_T + y_{To} - f} \\ z_m = -f\dfrac{z_C}{y_C - f} = -f\dfrac{r_{31}x_T + r_{32}y_T + r_{33}z_T + z_{To}}{r_{21}x_T + r_{22}y_T + r_{23}z_T + y_{To} - f} \end{cases} \tag{3}$$

where f stands for the focal length of the camera and $\{x_m, z_m\}^T$ denotes the projected image coordinates of the feature point. For a calibrated camera, the focal length is known in advance.

Defining $\eta = y_C - f$ in Eq. (3) leads to

$$\begin{cases} \eta x_m + x_C f = 0 \\ \eta z_m + z_C f = 0 \end{cases} \tag{4}$$

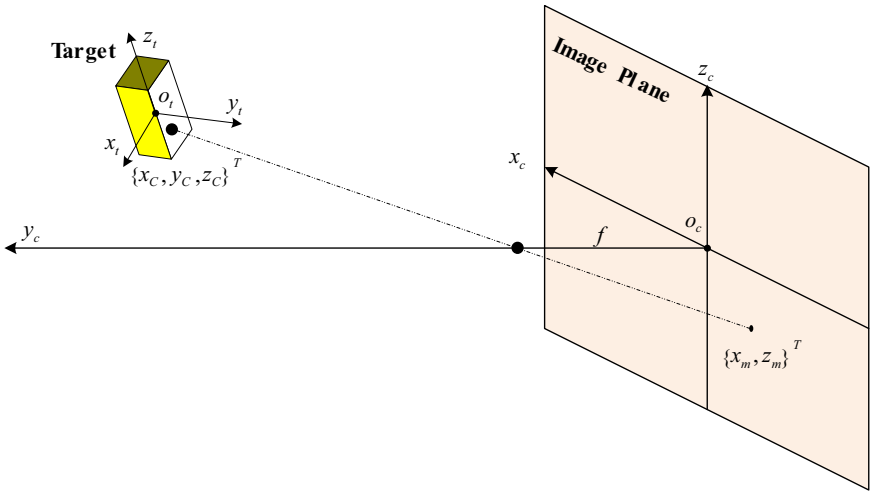

Fig. 1. Pin-hole camera model.

Further define the left side of Eq. (4) by $F=\eta x_m+x_C f$ and $G=\eta z_m+z_C f$, where F and G are functions of the target pose in the camera frame and the projected image coordinates of the feature point on the target. Linearizing F and G by the Taylor expansion in the vicinity of $\{x_m, z_m, x_{To}, y_{To}, z_{To}, \theta_x, \theta_y, \theta_z\}^T$ leads to

$$\boldsymbol{E}_{2\times1}+\boldsymbol{\delta}_{2\times1}=\boldsymbol{J_c}_{\,2\times6}\boldsymbol{dX_p}_{\,6\times1} \tag{5}$$

where

$$\boldsymbol{E}_{2\times1}=\begin{Bmatrix} dx_m \\ dz_m \end{Bmatrix},\quad \boldsymbol{\delta}_{2\times1}=\frac{1}{\eta}\begin{Bmatrix} F_0 \\ G_0 \end{Bmatrix},$$

$$\boldsymbol{dX_p}_{\,6\times1}=\{dx_{To}, dy_{To}, dz_{To}, d\theta_x, d\theta_y, d\theta_z\}^T,$$

$$\boldsymbol{J_c}_{\,2\times6}=-\frac{1}{\eta}\begin{Bmatrix} \left(\frac{\partial F}{\partial x_{To}}\right)_0 & \left(\frac{\partial F}{\partial y_{To}}\right)_0 & \left(\frac{\partial F}{\partial z_{To}}\right)_0 & \left(\frac{\partial F}{\partial \theta_x}\right)_0 & \left(\frac{\partial F}{\partial \theta_y}\right)_0 & \left(\frac{\partial F}{\partial \theta_z}\right)_0 \\ \left(\frac{\partial G}{\partial x_{To}}\right)_0 & \left(\frac{\partial G}{\partial y_{To}}\right)_0 & \left(\frac{\partial G}{\partial z_{To}}\right)_0 & \left(\frac{\partial G}{\partial \theta_x}\right)_0 & \left(\frac{\partial G}{\partial \theta_y}\right)_0 & \left(\frac{\partial G}{\partial \theta_z}\right)_0 \end{Bmatrix}.$$

Eq. (5) contains two independent equations for six unknowns (pose of target). Theoretically, one needs only three feature points to solve for the six unknowns. However, this approach may result in four ambiguous poses [28,29]. To eliminate the ambiguity and increase the robustness, minimum four feature points are widely adopted in literature, which leads to eight equations with six unknowns, such that

$$\boldsymbol{E}_{8\times1}+\boldsymbol{\delta}_{8\times1}=\boldsymbol{J_c}_{\,8\times6}\boldsymbol{dX_p}_{\,6\times1} \tag{6}$$

The unknowns are solved by an iterative least square approach assuming the zero residual error in the measurement, such that

$$\boldsymbol{E}_{8\times1}=0,\quad \boldsymbol{dX_p}_{\,6\times1}=(\boldsymbol{J_c}^{\boldsymbol{T}}_{\,8\times6}\boldsymbol{J_c}_{\,8\times6})^{-1}\boldsymbol{J_c}^{\boldsymbol{T}}_{\,8\times6}\boldsymbol{\delta}_{8\times1} \tag{7}$$

By inputting the known image coordinates of feature points and an initial guess of the target pose, the algorithm iterates to correct previous guess until the correction is less than a pre-set tolerance, i.e., $\|\boldsymbol{dX_p}\|\le\varepsilon$. In practice, it is common to use the previous target pose as the initial guess to reduce the iterations. As aforementioned, the photogrammetry is memoryless and prone to the image noise, which may result in large fluctuation of estimated target pose. Thus, the computational time of photogrammetry may increase if the initial guess, which is the previous pose as mentioned, is far away from the current pose. As a result, the sampling time-step of vision system may be adjusted, which is not desirable when dealing with real-time pose estimation. Another shortcoming of the photogrammetry is that it does not estimate the motion of target directly, which is an important parameter for trajectory planning of the robotic manipulator to perform autonomous capture in a dynamic environment. To address these challenges, an extended Kalman filter (EKF) with photogrammetry is presented in the following.

2.2. Extended Kalman filter

The Kalman filter is an optimal estimation algorithm for a linear system with independent white noise of normal distribution [41,42]. The camera model in Eq. (3) is highly nonlinear and the extended Kalman filter has been adopted to estimate the pose and motion of a dynamic target. Assume the motion of the target is approximately constant within the sampling interval if the sampling time-step is sufficiently small. Then, the motion of the target can be approximated by the first order equation of motion. Define the system variable vector as

$$\boldsymbol{X}=\{x_{To}, \dot{x}_{To}, y_{To}, \dot{y}_{To}, z_{To}, \dot{z}_{To}, \theta_x, \dot{\theta}_x, \theta_y, \dot{\theta}_y, \theta_z, \dot{\theta}_z\}^T$$

By assuming the system's acceleration vector $\boldsymbol{\omega}=\{\ddot{x}_{To}, \ddot{y}_{To}, \ddot{z}_{To}, \ddot{\theta}_x, \ddot{\theta}_y, \ddot{\theta}_z\}^T$ as the process noise with a normal distribution with zero mean and covariance matrix $\boldsymbol{Q}$, the system model is expressed as

$$\boldsymbol{X}_k=\boldsymbol{A}\boldsymbol{X}_{k-1}+\boldsymbol{B}\boldsymbol{\omega}_{k-1} \tag{8}$$

where

$$\boldsymbol{A}=diag\begin{bmatrix}\boldsymbol{A_1} & \boldsymbol{A_1} & \boldsymbol{A_1} & \boldsymbol{A_1} & \boldsymbol{A_1} & \boldsymbol{A_1}\end{bmatrix},\quad \boldsymbol{A_1}=\begin{bmatrix}1 & t \\ 0 & 1\end{bmatrix},$$

$$\boldsymbol{B} = diag\left[\boldsymbol{B_1} \quad \boldsymbol{B_1} \quad \boldsymbol{B_1} \quad \boldsymbol{B_1} \quad \boldsymbol{B_1} \quad \boldsymbol{B_1}\right], \quad \boldsymbol{B_1} = \begin{Bmatrix} t^2/2 \\ t \end{Bmatrix}.$$

k is sample time-step and t is the sample time.

The measurement model is developed from the pin-hole camera model in Eq. (3) for n feature points, such that

$$\begin{aligned} &\boldsymbol{Z}_k = \boldsymbol{h}(\boldsymbol{X}_k) + \boldsymbol{\mu}_k \\ &\boldsymbol{Z} = \{x_{m1}, z_{m1}, \cdots x_{mn}, z_{mn}\}^T, \quad \boldsymbol{h}(\cdot) = \{h_{x1}, h_{z1}, \cdots h_{xn}, h_{zn}\}^T \end{aligned} \tag{9}$$

and

$$h_{xi}(\boldsymbol{X}) = -f\frac{x_{Ci}}{\eta_i}, \quad h_{zi}(\boldsymbol{X}) = -f\frac{z_{Ci}}{\eta_i} \tag{10}$$

where $\boldsymbol{\mu}$ is the measurement noise that obeys a normal distribution with zero mean and covariance matrix $\boldsymbol{R}$.

The EKF requires initial conditions and measurements observed over time. Since initial conditions of a non-cooperative target are unknown, an inappropriate initial guess may lead to poor performance of the EKF. To improve its performance and accelerate the convergence rate, we initialize the state variable vector by the photogrammetry in the algorithm. Based on the above definitions, a recursive pose and motion estimation algorithm is derived as shown in Table 1, where $\boldsymbol{H}$ is the Jacobian matrix formed by the first order partial differential of the measurement model respect to system variable, $\boldsymbol{K_g}$ is the Kalman gain at time step k, $\boldsymbol{P}$ is the covariance matrix of the system state variable, $\boldsymbol{Q}$ and $\boldsymbol{R}$ are the process and measurement noise covariance matrices.

2.3. Kinematics of robotic manipulator

The autonomous capture will be conducted by a custom-built six degrees of freedom (6DOF) robotic manipulator with an eye-in-hand configuration as shown in Fig. 2. The robotic manipulator consists of three links and one end-effector with five revolute and one prismatic joints. The eye-in-hand camera is mounted closely to the end-effector. The first three revolute joints, namely torso (θ_1), shoulder (θ_2) and elbow (θ_3), control the position of the end-effector while the last two revolute joints and one prismatic joint, namely wrist roll (θ_4), wrist yaw (θ_5) and gripper (d_6), provide dexterous orientation and griping function for capture operation. Thus, the translation and rotation of the end-effector can be considered separately from the wrist to simplify the controller design.

The kinematics of robotic manipulator provides the forward relationship from the joint angles to the position of the end-effector (the wrist center), such that

$$\boldsymbol{X_g} = \boldsymbol{K_f}(\boldsymbol{\Theta}) \tag{11}$$

where $\boldsymbol{\Theta} = \{\theta_1, \theta_2, \theta_3\}^T$.

Table 1
Pose and motion estimation algorithm.

1.	Given an initial guess of target pose: $\boldsymbol{X}_p = \boldsymbol{X}_{p0}$;
2.	Input measurement;
3.	Start photogrammetry loop
4.	{
5.	evaluate pin-hole camera model Jacobian $\boldsymbol{J}_c$ and image error vector $\boldsymbol{\delta}$;
6.	calculate pseudo-inverse of Jacobian: $\boldsymbol{J}_c{}^+ = \left(\boldsymbol{J}_c{}^T\boldsymbol{J}_c\right)^{-1}\boldsymbol{J}_c{}^T$;
7.	calculate previous-guess-correction: $\boldsymbol{dX}_p = \boldsymbol{J}_c{}^+\boldsymbol{\delta}$;
8.	if ($\|\boldsymbol{dX}_p\| <$ tolerance)
9.	{break;}
10.	else
11.	$\{\boldsymbol{X}_p = \boldsymbol{X}_p + \boldsymbol{dX}_P;\}$
12.	}
13.	Return $\boldsymbol{X}_p$;
14.	Augment to state variable: $\boldsymbol{X}_0 = \left(\boldsymbol{X}_p, \dot{\boldsymbol{X}}_p\right)$;
15.	Initialize EKF: $\boldsymbol{X}_0$ (initial state variable), $\boldsymbol{P}_0$ (state variable covariance matrix), $\boldsymbol{Q}$ (process noise covariance matrix), $\boldsymbol{R}$ (measurement noise covariance matrix);
16.	Start EKF loop
17.	{
18.	estimate next state variable and covariance matrix: $\boldsymbol{X}_{k\|k-1} = \boldsymbol{A}\boldsymbol{X}_{k-1\|k-1}$ $\boldsymbol{P}_{k\|k-1} = \boldsymbol{A}\boldsymbol{P}_{k-1\|k-1}\boldsymbol{A}^T + \boldsymbol{B}\boldsymbol{Q}\boldsymbol{B}^T$
19.	evaluate Jacobian of measurement model: $\boldsymbol{H}_k = \partial\boldsymbol{h}(\boldsymbol{X})/\partial\boldsymbol{X}\|_{\boldsymbol{X} = \boldsymbol{X}_{k\|k-1}}$
20.	calculate Kalman gain: $\boldsymbol{K}_g = \boldsymbol{P}_{k\|k-1}\boldsymbol{H}_k^T\left(\boldsymbol{H}_k\boldsymbol{P}_{k\|k-1}\boldsymbol{H}_k^T + \boldsymbol{R}\right)^{-1}$
21.	update state variable and covariance matrix: $\boldsymbol{X}_{k\|k} = \boldsymbol{X}_{k\|k-1} + \boldsymbol{K}_g\left[\boldsymbol{Z}_k - \boldsymbol{h}\left(\boldsymbol{X}_{k\|k-1}\right)\right]$ $\boldsymbol{P}_{k\|k} = \boldsymbol{P}_{k\|k-1} - \boldsymbol{K}_g\boldsymbol{H}_k\boldsymbol{P}_{k\|k-1}$
22.	record and output: $\boldsymbol{X}_{k\|k}$, $\boldsymbol{P}_{k\|k}$;
23.	}

Thus, the velocity of the end-effector can be derived by taking the time derivative of Eq. (11)

$$\dot{\boldsymbol{X}}_g = \boldsymbol{J}_r \dot{\boldsymbol{\Theta}} \tag{12}$$

where $\boldsymbol{J}_r$ is the Jacobian matrix of the robotic manipulator formed by joint angles.

Eqs. (11) and (12) form the forward kinematics of the robotic manipulator. The actuators are step motors at joints, which drive the end-effector to the desired position by achieving the desired joint angles. Thus, the transformation of position to joint angles, namely inverse kinematics, should be derived in advance. It is well known that the robotic inverse kinematics may have multiple-solutions due to the periodicity of trigonometric functions. In order to obtain a unique solution of joint angles based on Eq. (11), we introduced additional constraints, such as mechanical configuration and motion limits as per [16] in addition to the following inverse kinematics

$$\boldsymbol{\Theta} = \tilde{\boldsymbol{K}}_f^{\,-1} \boldsymbol{X}_g, \quad \dot{\boldsymbol{\Theta}} = \boldsymbol{J}_r^{\,-1} \dot{\boldsymbol{X}}_g \tag{13}$$

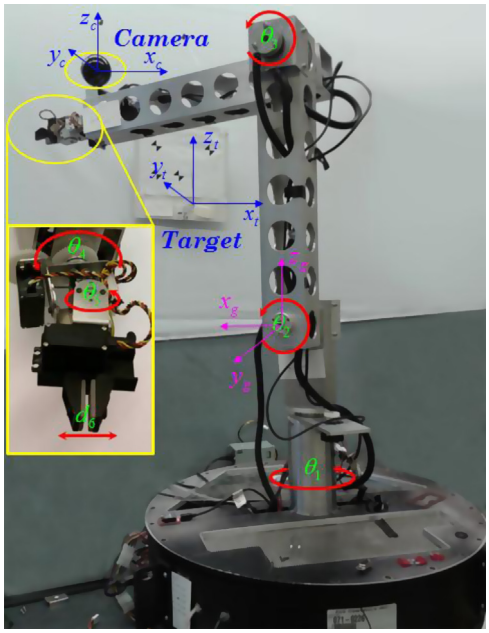

Fig. 2. The custom-built 6DOF robotic manipulator.

2.4. Desired state of the end-effector

The desired state of the end-effector is defined as the pose and motion of a target in the dynamic PBVS. Since the pose and motion of a target are estimated relative to the camera frame, they must be transformed to the global frame first, such that

$$\boldsymbol{X}_g = \boldsymbol{T}_{cg} \boldsymbol{X}_c \tag{14}$$

$$\dot{\boldsymbol{X}}_g = \dot{\boldsymbol{T}}_{cg} \boldsymbol{X}_c + \boldsymbol{T}_{cg} \dot{\boldsymbol{X}}_c \tag{15}$$

where $\boldsymbol{T}_{cg}$ is the transformation matrix from the camera frame to the global frame.

Denote the position and velocity vectors of a target as

$$\boldsymbol{X}_c^t = \{x_{To}, y_{To}, z_{To}\}^T, \quad \dot{\boldsymbol{X}}_c^t = \{\dot{x}_{To}, \dot{y}_{To}, \dot{z}_{To}\}^T \tag{16}$$

Then, the desired position and velocity of the end-effector in the global frame are determined by substituting Eqs. (14) and (15) into Eq. (16)

$$\boldsymbol{X}_g^d = \boldsymbol{T}_{cg} \boldsymbol{X}_c^t, \quad \dot{\boldsymbol{X}}_g^d = \dot{\boldsymbol{T}}_{cg} \boldsymbol{X}_c^t + \boldsymbol{T}_{cg} \dot{\boldsymbol{X}}_c^t \tag{17}$$

Accordingly, the desired joint angles and angular velocities of the first three joints of the robotic manipulator are determined by substituting Eq. (17) into Eq. (13), such as

$$\left\{ \theta_1^d \quad \theta_2^d \quad \theta_3^d \right\}^T = \tilde{\boldsymbol{K}}_f^{\,-1} \boldsymbol{X}_g^d, \quad \left\{ \dot{\theta}_1^d \quad \dot{\theta}_2^d \quad \dot{\theta}_3^d \right\}^T = \boldsymbol{J}_r^{\,-1} \dot{\boldsymbol{X}}_g^d \tag{18}$$

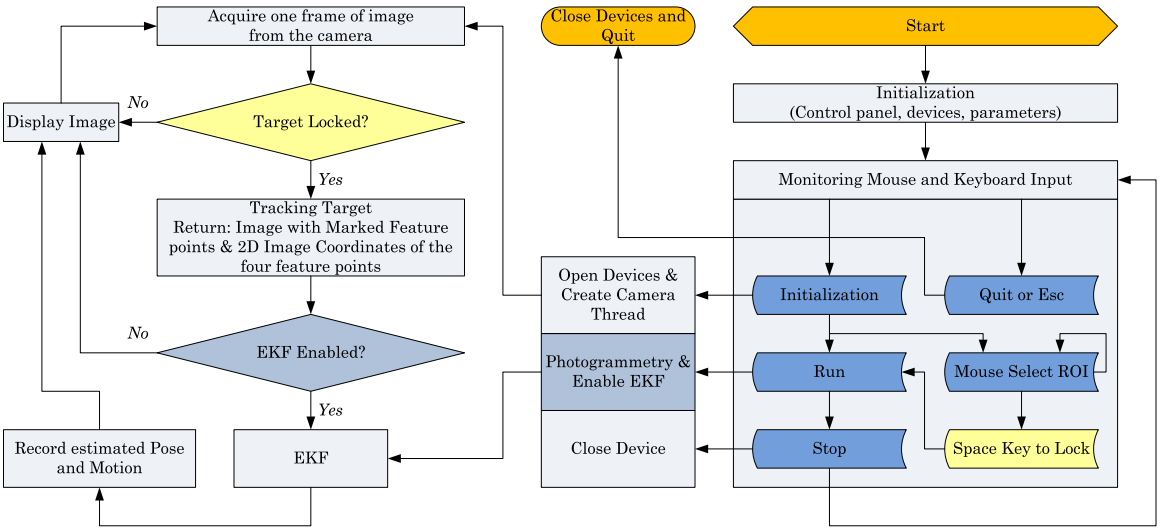

Fig. 3. Flowchart of the pose and motion estimation algorithm.

The desired joint angles and angular velocities of the wrist roll and wrist yaw are defined as

$$\left\{ \theta_4^d \quad \theta_5^d \quad \dot{\theta}_4^d \quad \dot{\theta}_5^d \right\} = \left\{ \theta_y \quad \theta_z \quad \dot{\theta}_y \quad \dot{\theta}_z \right\} \tag{19}$$

They will be determined separately by the capture strategy defined below.

2.5. Capture strategy

Once the target's pose and position are estimated by the EKF, the task of autonomous tracking, approaching and capturing of the target will be performed by an inner control loop of the robot. For the dynamic capture, the robotic manipulator should track and approach the interception point of the moving target instead of its current position [16], which is approximately estimated as

$$\boldsymbol{\Theta}^{\boldsymbol{I}} = \boldsymbol{\Theta}^{\boldsymbol{d}} + \dot{\boldsymbol{\Theta}}^{\boldsymbol{d}}(\Delta T + t) \tag{20}$$

where t is the sample time of the vision system and $\Delta T = \left(\|\boldsymbol{X}_{\boldsymbol{g}}^{\boldsymbol{d}} - \boldsymbol{X}_{\boldsymbol{g}}\| / \|\dot{\boldsymbol{X}}_{\boldsymbol{g}}\| \right) = \left(\|\boldsymbol{T}_{\boldsymbol{cg}} \boldsymbol{X}_{\boldsymbol{c}}^{\boldsymbol{t}} - \boldsymbol{K}_{\boldsymbol{f}}\| / \|\boldsymbol{J}_{\boldsymbol{r}} \dot{\boldsymbol{\Theta}}\| \right)$ is the estimated time for the end-effector to approach the desired position from its current position at its maximum allowed velocity. It should be noted that the purpose of time ΔT is to provide a lead-time for the controller to estimate the interception point and the accuracy of its estimate is not critical. This is because ΔT as well as the error in the estimation of interception point in Eq. (20) will be reduced as the robot approaches the target.

Thus, the control for joint actuators with respect to their current positions is obtained as

$$\Delta \boldsymbol{\Theta}^C = \boldsymbol{\Theta}^d + \dot{\boldsymbol{\Theta}}^d(\Delta T + t) - \boldsymbol{\Theta} \tag{21}$$

Once the target is within the capture region of the end-effector, the control input for the wrist roll and yaw angles is the errors between the orientations of the griper and the target, which can be simply defined as

$$\Delta \theta_{4,5}^C = \Delta \theta_{4,5}^d + \dot{\theta}_{4,5}^d t \tag{22}$$

The control strategy to activate a capture is given in [16].

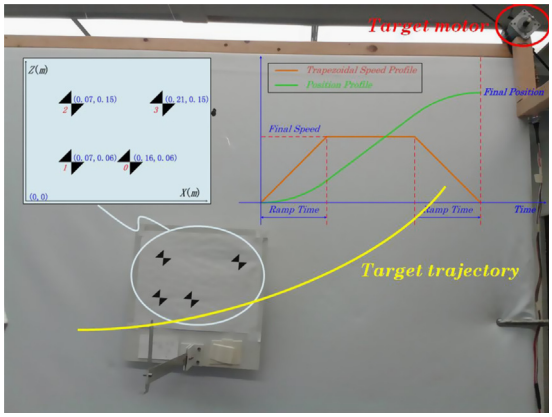

Fig. 4. Non-cooperative target system.

3. Experimental setup

3.1. Robotic manipulator system

The proposed pose and motion estimation algorithm for the non-cooperative target and the PBVS autonomous capture by the robotic manipulator are validated

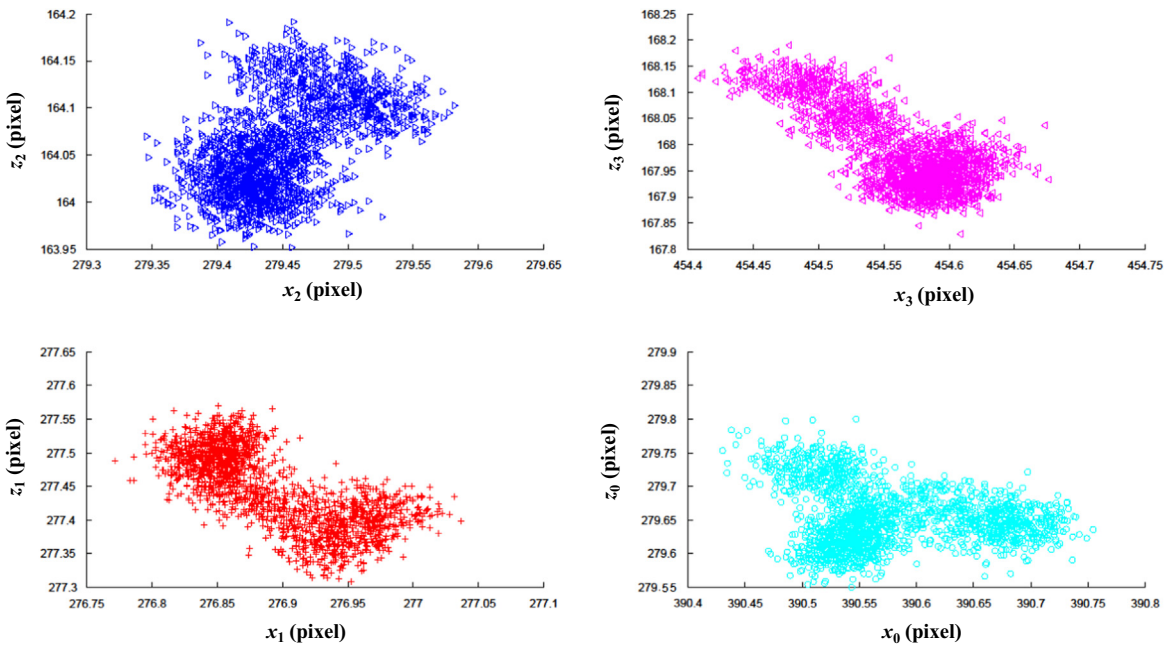

Fig. 5. The centroid coordinates of feature points of a stationary target on image plane.

experimentally. A robotic manipulator with five revolute joints, one prismatic joint and an eye-in-hand camera mounted closely to the end effector is custom built to conduct the autonomous PBVS capture operation, as shown in Fig. 2. The camera used in the experiment was a Logitech webcam with 640×480 resolution and 2 mm focal length. The implementation of the pose and motion estimation and PBVS control has been done by the Microsoft Fundamental Class (MFC) in order to obtain a friendly human machine interface. The flowchart of the implementation is shown in Fig. 3. Once the camera is activated, a separate thread is created simultaneously for the camera to acquire image data and display them on the computer screen. After the target is locked, the imaging processing subroutine is activated to track the motion of feature points and output their pixel coordinates in sequence.

3.2. Non-cooperative target system

A target with a known pattern of four low-optical-noise features is designed to facilitate a simple optical locking and tracking. The target system as well as configuration of feature points is shown in Fig. 4. The target is driven by a single stepper motor with programmable speed profile in the tests. By programming the speed profile of the motor and isolating the target system from the robot system, several testing scenarios for the non-cooperative target can be generated.

The task of imaging processing is to track the center point of four optical features on the target. It is done by the Open-CV (Open Source Computer Vision) software. By applying the *cvGoodFeaturesToTrack* and *cvFindCornerSubPix* functions of OpenCV library, the target tracking program groups the corners of each project feature image to extract the centroid coordinates and tracks the centers of each group by *cvCalcOpticalFlowPyrLK* function. The extracted coordinates of the center of each feature of a stationary target are shown in Fig. 5. It can be seen that the image noises of each feature points are within an area of 0.5×0.5 pixel due to jittering. Thus, the covariance matrix of the measurement noise in EKF can be determined accordingly as, $\boldsymbol{R} = e^2 \boldsymbol{I}_{8\times 8}$, where e stands for the metric dimension of half-pixel. The covariance matrix of process noise, $\boldsymbol{Q} = 10^{-5} \boldsymbol{I}_{6\times 6}$, was determined by trial-and-error.

4. Experimental results and discussion

4.1. Static pose estimation by EKF

The target was stationary in the global frame in the static pose estimation. It was placed roughly 0.5 m away from the camera. The pose estimates by the EKF, as shown in Fig. 6, are compared with the results by photogrammetry. As expected, the results of photogrammetry fluctuate around the true pose of the target due to the jittering of pixel coordinates of features. The jitter distribution in the experiments was observed as ± 0.5 pixel as shown in Fig. 5. It resulted in roughly 0.1 mm translational and 0.4° rotational fluctuations in pose estimation as shown in Fig. 6(a). This is because the photogrammetry does not consider the measurement noises of images and propagates the noises down to the pose estimation. In contrast, the poses estimated by the EKF are much smoother than the photogrammetry. The EKF suppressed the spread of the measurement noises by considering the effects of measurement noises. The motion estimated by the EKF is stationary compared with the one estimated based on the numerical derivation of the photogrammetry data, which is very noisy and unsuitable for robotic control, see Fig. 6(b). This experiment demonstrates the necessity of using the EKF in the PBVS of robots.

4.2. Dynamic pose and motion estimation by EKF

The dynamic pose and motion estimation was done by manually translating and rotating the target along x, y, z axes, while the target was roughly 0.5 m away from the camera. The estimated pose and motion of the moving target are shown in Fig. 7 together with the results from the photogrammetry. Fig. 7(a) shows that the positions estimated by the EKF and photogrammetry are in good agreement. However, the orientations estimated by the photogrammetry are noisier than the EKF. It is worth pointing out that the EKF requires the Jacobian matrix of measurement model, which induces linearization error. The stronger the nonlinearity, the more error may be induced. According to Eq. (3), the nonlinearity of y, θ_x, θ_z are relatively stronger than x, z, θ_y in the measurement model. Therefore, it can be seen the test results of y, θ_x, θ_z estimated by the EKF agree with the photogrammetry better than x, z, θ_y. In the latter case, the estimates by the EKF show some delay in phase when the target experienced sudden motion. Fig. 7(b) shows the motion estimated by the EKF. As a comparison, the motion obtained by a simple numerical derivation of the photogrammetry data is also plotted. It shows clearly that the EKF results are much smoother than the photogrammetry and suitable for robotic tracking control.

4.3. Robotic autonomous capture enhanced by EKF

Once the EKF algorithm is validated, the robotic autonomous capture strategy is tested using the custom-built robotic manipulator. The robotic manipulator always starts at its home position, i.e., torso 0°, shoulder 90° and elbow 0°. The speed profile of the target motor is a trapezoidal with a ramp of two seconds in the dynamic capture testing. The maximum linear velocity of the target was set to 1.2 cm/s and the direction of the velocity vector changed in the course. Fig. 8 shows the test results where the autonomous capture process was divided into three phases. Phase 1 is the target searching and locking until the target was locked by the vision system. In phase 2, the vision system tracked the motion of the target using optical flow to increase the imaging process efficiency. The robot started to track and approach the target based on the estimated information of pose and motion of the target. The desired and actual joint angles change simultaneously as the end-effector approaches the target. This is because the position of the target with respect to the eye-in-hand camera varied continuously when the pose error between the end-effector and the target diminished.

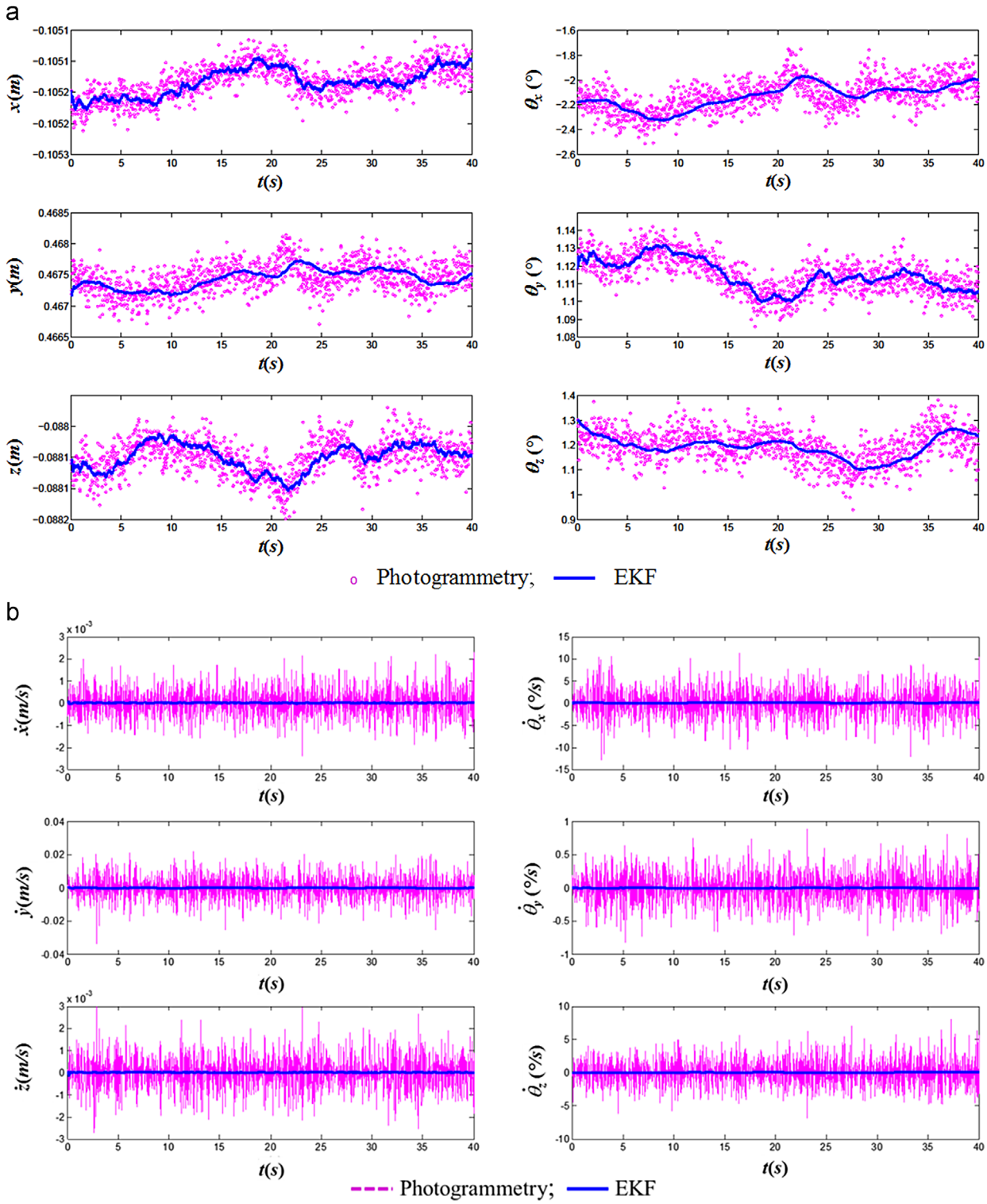

Fig. 6. Static test results: (a) pose estimation and (b) motion estimation.

However, the estimated desired joint angles were smoother than the actual joint angles due to some residual vibration in joints. Furthermore, the target motion caused the estimated desired joint angles to vary ahead of the actual joint angles as shown in Fig. 8 because the end-effector was approaching to the interception instead of the current target position. Finally in phase 3, the manipulator successfully captured the moving target.

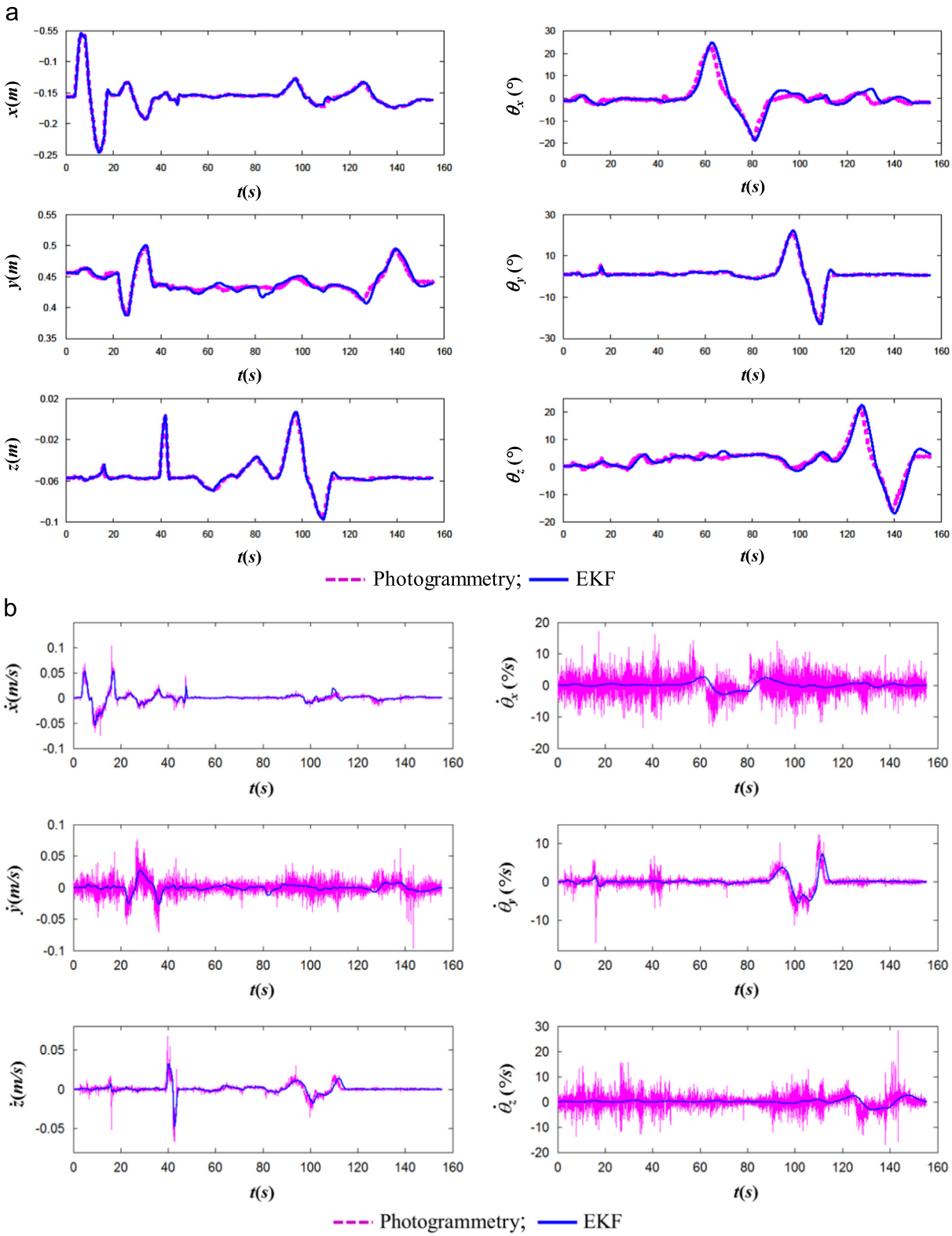

Fig. 7. Dynamic test results: (a) pose estimation and (b) motion estimation.

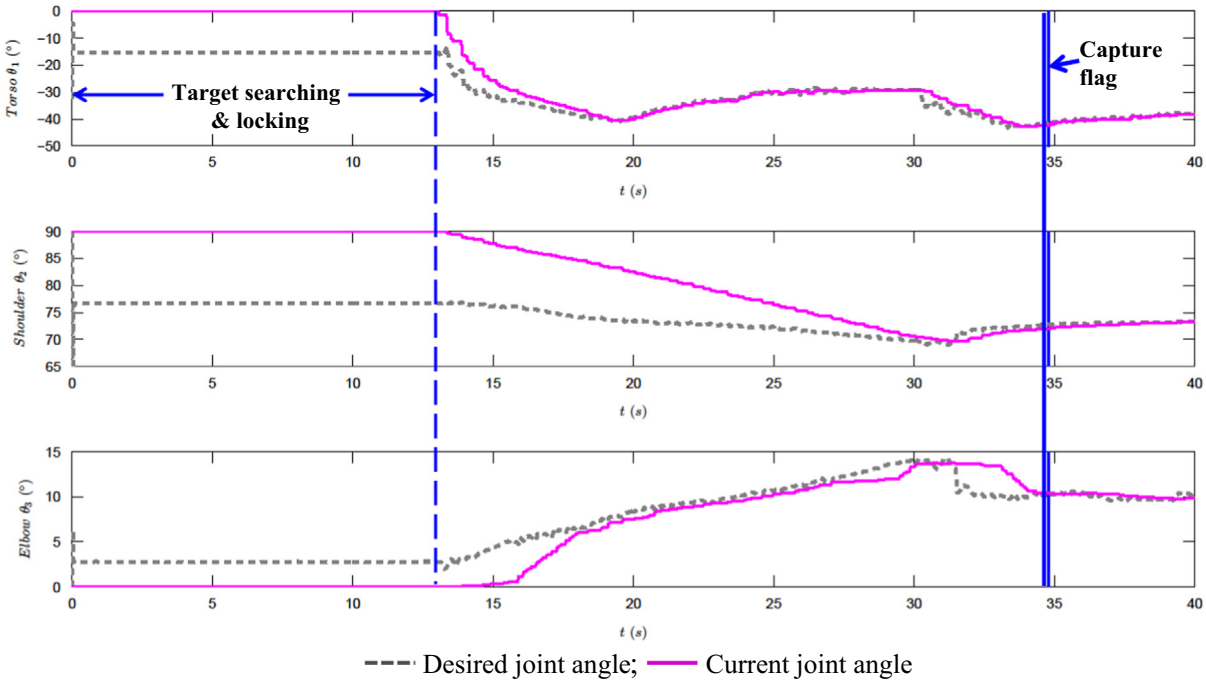

Fig. 8. Test results of autonomous capture of moving target.

5. Conclusion

This paper proposed a real-time PBVS control of autonomous robotic capture of a non-cooperative target. A vision-based pose and motion estimation algorithm of a non-cooperative target was developed by the photogrammetry and the extended Kalman filter for robotic manipulators to perform the PBVS autonomous capture. This methodology adopts the photogrammetry to initialize the EKF to improve its convergence rate when dealing with the non-cooperative target case. Optical flow algorithm is employed to track the target in order to increase the imaging processing speed for real-time pose and motion estimation of the non-cooperative target. This is beneficial for reducing the time delaying in joint actuation. A close-loop PBVS control strategy is devised to determine the desired pose and position of the end-effector at the rendezvous point once the pose and motion of the target are estimated. The proposed approach was validated experimentally on a custom-built robotic manipulator with an eye-in-hand configuration. The experimental results show that the EKF provided a smooth estimation of pose and motion of the target and enabled successful captures of a non-cooperative target. The experiments demonstrated that the effectiveness and robustness of the proposed EKF enhanced pose and motion estimation and the PBVS control strategy.

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